

Chapter: Stone Game III (LC-1406) Notes

Authored by: Arpit Raj, LNMIIT Jaipur | August 21, 2026

Algorithm Notes & Logic

DP Array Definition

`dp[i][playernum]` : If 0 then *aadz*, if 1 then *bob*.

dp best score dega

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State Transitions

agar ith index ka uthaye to `i+x` tak ja skte.

`currscore = took + dp[i+x][1]` , since depends on future so top down.

x can be 1, 2, 3.

Variables Breakdown

min of 3 or number of piles left.

n is total piles, i is pile number we oin.

Implementation Code

```

class Solution {
public:
    string stoneGameIII(vector<int>& stonevalue) {

        int n=stonevalue.size();
        vector<int> pref(n+1,0);

        for(int i=1;i<=n;i++){
            pref[i]=pref[i-1]+stonevalue[i-1];
        }

        //dp[i][playernum]
        //if 0 then aadz
        //if 1 then bob
        //dp best score dega
        vector<vector<int>> dp(n+1,vector<int>(2,0));

        //agar ith index ka uthaye to i+x tak ja skte
        //currscore=took+dp[i+x][1] , since depends on future so top down
        //x can be 1,2,3

        for(int i=n-1;i>=0;i--){
            int bestcurrscore=INT_MIN;
            int bestotherscore=INT_MIN;

            for(int x=1;x<=min(3,n-i);x++){ //min of 3 or number of piles left
                //n is total piles , i is pile number we o

                int took=pref[i+x]-pref[i];
                int currscore=took+dp[i+x][1];
                int otherscore=dp[i+x][0];

                if(currscore>bestcurrscore){
                    bestcurrscore=currscore;
                    bestotherscore=otherscore;
                }
            }

            dp[i][0]=bestcurrscore;
            dp[i][1]=bestotherscore;

        }

        int aadyasc=dp[0][0];
        int mysc=dp[0][1];

        if(aadyasc>mysc){
            return "Alice";
        }
        else if(aadyasc<mysc){
            return "Bob";
        }

        else{
            return "Tie";
        }
    }
};

```

```
}  
  
}  
};
```